Factor-Anchored Reinforcement Learning for Portfolio Risk Control

Evidence from Frozen-Bundle Evaluation under Timing and Cost Constraints

Ioan Hedea

Preprint prepared in a NeurIPS-style conference template.

Abstract

This paper studies whether reinforcement learning is more useful in equity trading as a control layer than as an end-to-end predictor. The evaluated setup is a modular, factor-anchored pipeline in which finance-first alpha models feed a constrained allocator, while RL modulates exposure, protection, and execution under explicit timing and transaction-cost assumptions. Rather than relying on a best-case exploratory run, the paper ties its claims to a frozen archived bundle spanning March 2021 to March 2026. In that archived evaluation, the full pipeline improves maximum drawdown by 8.6 percentage points relative to SPY (-10.2% versus -18.8%) and exceeds an end-to-end PPO baseline by 8.1 percentage points of annualized return (17.2% versus 9.1%). However, the factor-only benchmark and simple rule-based overlays still achieve materially higher Sharpe ratios, around 1.9 versus 1.23 for the full RL stack. The supported conclusion is therefore deliberately narrow: RL adds value here primarily through interpretable downside control rather than through top-line performance leadership.

Keywords: quantitative trading, reinforcement learning, factor investing, statistical arbitrage, regime detection, risk management, backtesting

1 Introduction

Many quantitative trading systems are built from a collection of specialized modules rather than from a single monolithic predictor. Cross-sectional factors, relative-value signals, volatility models, and regime filters often contribute distinct pieces of information about expected return and risk. Reinforcement learning (RL), by contrast, is most compelling when the task is inherently sequential: choosing when to increase risk, when to hedge, and how to adapt execution to changing conditions [1, 23].

This project adopts exactly that separation of responsibilities. The alpha layer is intentionally finance-first. It draws on ideas from factor investing [10, 15], cointegration-based statistical arbitrage [9], conditional volatility modeling [3, 8], and regime detection [13]. The RL layer does not attempt to discover alpha from scratch. Instead, it treats alpha as an input and learns how aggressively to express it under different risk and market states.

The present implementation contributes five practical elements to the evaluation setup.

1. A modular code architecture that cleanly separates data, alpha, control, backtesting, and visualization concerns.
2. A factor-anchored portfolio construction policy in which RL primarily controls exposure and active overlay size rather than acting as a second stock selector.
3. A constrained optimization layer between alpha and RL that turns signal forecasts into a capped, turnover-aware, and group-aware target book.

4. A light dynamic hedging sleeve and transaction-cost model that make the backtest materially more realistic without requiring an options chain or intraday execution feed.
5. A research evaluation workflow including ablations, rolling windows, sensitivity tests, and regime-conditioned policy summaries.

1.1 Research Questions

The current version of the report is organized around three explicit research questions.

1. **RQ1: Controller vs. end-to-end RL.** Is reinforcement learning more effective as a modular control layer over financially grounded alpha models than as an end-to-end trading agent?
2. **RQ2: Value over simple rules.** Does the RL control layer add measurable value beyond rule-based heuristics such as volatility targeting and drawdown-based deleveraging?
3. **RQ3: Component attribution.** Which part of the control layer, exposure modulation, dynamic hedging, or their joint operation, is the primary driver of the observed risk transformation?

Beyond the Present Manuscript

A natural next step, not evaluated in the present manuscript, concerns hedge realism: whether implied-volatility-conditioned, option-style overlays such as protective puts, collars, and put spreads improve the downside sleeve beyond the current stylized convex hedge approximation.

From a research perspective, this framing matters because it narrows the gap between a teaching prototype and a system that can be discussed in the language of quantitative portfolio construction. The objective is not to claim that the pipeline is production-ready. Rather, it is to assemble a well-motivated stack in which each modeling choice has a clear financial role and where performance can be decomposed into alpha generation, risk transformation, and control.

The central defended statement of the paper is therefore sharper than "the system works." *A factor-anchored RL control layer, evaluated under explicit cost and timing assumptions, can improve downside control relative to passive exposure while remaining more interpretable than an end-to-end policy. The paper does not claim that this RL layer surpasses the strongest finance-first or rule-based baselines on return- and Sharpe-based comparisons.* That narrower empirical outcome is still useful, because it helps separate where RL contributes to risk transformation from where simpler finance-first mechanisms remain stronger. The rest of the paper is organized to test that claim rather than merely to catalogue modules.

2 Related Literature and Design Motivation

The report is not meant to claim novelty in any individual modeling choice. Instead, it builds a coherent pipeline from well-established ingredients.

Factor investing. The cross-sectional component is aligned with the idea that common risk factors help explain returns [10]. Momentum is motivated by the medium-horizon continuation evidence of Jegadeesh and Titman [15], while quality and low-volatility proxies are included to stabilize the cross-section.

Statistical arbitrage. The pairs sleeve follows the classical cointegration logic of Engle and Granger [9]: if a spread is mean reverting around an equilibrium relation, deviations from that equilibrium may carry predictive content.

Volatility and regimes. Conditional heteroskedasticity motivates the GARCH sleeve [3, 8], while the

market-regime component uses a two-state hidden Markov model in the spirit of Hamilton [13].

Reinforcement learning. RL is used as a controller rather than a raw alpha generator. This follows the practical observation that in finance, RL is often most useful for allocation, inventory, and execution decisions rather than for replacing domain-specific signal construction [1, 23]. This role assignment is also motivated by the low signal-to-noise ratio and limited effective sample size typical of financial data, which make end-to-end policy learning especially brittle.

Rule-based risk management and controller baselines. A key part of the evaluation design is to compare the RL controller not only with passive benchmarks but also with rule-based heuristics that a skeptical reader would reasonably propose as simpler substitutes. In this project, the most important examples are volatility targeting and drawdown-based deleveraging. These baselines matter because they help distinguish "RL adds value" from "any reasonable overlay would have helped."

2.1 Recent Work on RL for Portfolio Management

Recent work on RL in portfolio management and trading can be organized into four overlapping strands.

End-to-end policy learning. Early and influential work uses RL to learn trading or allocation policies directly from price histories and realized rewards [5, 6, 16, 19, 21, 24]. The appeal of this approach is conceptual elegance: the same learner is asked to extract signal, size positions, and adapt sequentially. The difficulty is that financial returns are noisy, nonstationary, and data-poor relative to many other machine-learning domains. As a result, end-to-end RL systems can be sample-inefficient, regime-sensitive, and hard to interpret when they fail [12, 14, 17].

Neural return forecasting with a separate optimizer. A second line of work uses deep or other machine-learning models to estimate returns, factors, or pricing kernels, then hands those forecasts to a conventional optimizer [4, 11, 12]. This preserves more structure in the allocation step than end-to-end RL, and often performs well empirically, but it still concentrates modeling risk in a data-hungry prediction layer.

RL for execution, hedging, and risk control. A third strand applies RL to narrower sequential tasks such as execution, hedging, and dynamic risk management rather than to raw alpha discovery [2, 7, 18, 20]. This decomposition is closest to the present paper. The core idea is that RL may be most useful where the problem is inherently sequential and control-like, while alpha generation remains anchored to domain models or explicit financial structure.

Benchmarking and reproducibility. More recent platform and survey work has emphasized that RL in finance still suffers from weak evaluation discipline, narrow datasets, and inconsistent reporting [14, 22]. That critique directly motivates the frozen-bundle methodology used here. The paper is not only a claim about architecture; it is also a claim about how such architectures should be evaluated.

Table 1 positions the present approach relative to these broader families. The contribution is not a new RL algorithm. It is a methodological proposal: if strong finance-first alpha already exists, RL should be tested primarily as a controller layered on top of that alpha, then judged against both simple rule-based overlays and end-to-end learned alternatives.

Table 1: Taxonomy of portfolio-management architectures.

Approach	Alpha source	Control logic	Interp.	Representative references
Classical factors	Hand-crafted financial signals	Fixed allocator or simple overlay rules	High	Fama–French style factor investing [10]
End-to-end RL	Learned from market histories	RL policy directly selects positions or weights	Low to medium	Direct trading and portfolio RL [5, 16, 19, 24]
Neural forecasting + optimizer	Neural or ML return forecasts	Separate optimizer translates forecasts into holdings	Medium	Asset-pricing and return-forecast models [4, 11, 12]
This paper	**Finance-first alpha models**	**RL modulates exposure, hedging, and execution**	**High**	**Frozen-bundle control-layer evaluation**

2.2 Contribution Type

The contribution is not novelty in any single alpha ingredient. The paper does not claim a new factor model, a new GARCH variant, or a new hidden-state estimator. Its contribution lies instead in three places. First, it proposes a clear role assignment between alpha and control: financially grounded models produce expected-return structure, while RL handles sequential exposure and protection decisions. Second, it implements that role assignment in a modular architecture with an explicit constrained allocator between alpha and control. Third, it pairs the architecture with an evaluation workflow designed to test that claim via ablations, timing-discipline checks, robustness slices, and regime-conditional policy analysis. The contribution is therefore chiefly methodological: the paper proposes and tests a disciplined way to evaluate RL as a control layer rather than as a monolithic trading policy.

3 System Architecture

The pipeline is organized as a linear sequence of stages, illustrated in Figure 1. Each stage has a clearly delimited responsibility: upstream stages produce signals and risk features, the control layer translates these into portfolio decisions, and the backtest layer evaluates realized outcomes.

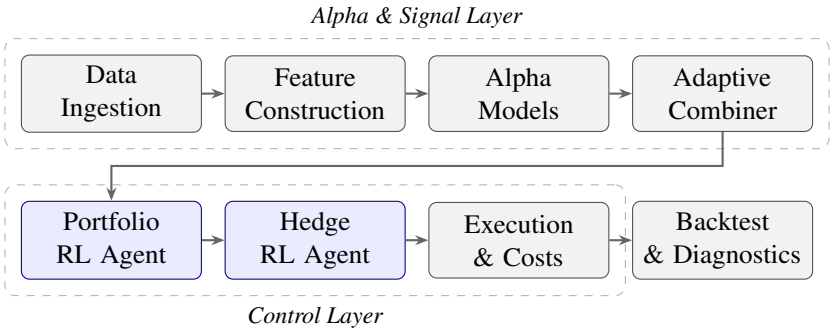

Figure 1: High-level architecture of the quantitative trading pipeline. The alpha and signal layer (top) produces expected-return scores and risk features; the control layer (bottom) translates these into portfolio weights, hedge intensity, and cash allocation before evaluation in the backtest engine.

3.1 Software Structure

Table 2: Module-level decomposition of the `quant_stack` package.

Module	Responsibility
`config.py`	Universe definition, global constants, and external series identifiers
`data.py`	Market price/volume ingestion, macro series (FRED / BLS fallback), and SEC company facts
`alpha.py`	Cross-sectional factors, pairs cointegration, GARCH volatility, HMM regime, LSTM forecast, and adaptive signal combination
`rl.py`	Portfolio construction agent, execution controller, and dynamic hedging agent
`pipeline.py`	Walk-forward backtest orchestration and online RL update loop
`evaluation.py`	Ablation, rolling-window robustness, sensitivity analysis, and research artifact export
`plots.py`	Equity-curve, drawdown, RL behavior, and alpha-model diagnostic figures
`main.py`	CLI entry point and compatibility wrapper

This decomposition is important for research reproducibility. It keeps the external command stable while turning the internals into separately testable components.

4 Data and Feature Construction

4.1 Universe

The traded universe spans four broad buckets. The growth sleeve comprises AAPL, MSFT, GOOGL, AMZN, META, and NVDA. Financials and energy are represented by JPM, GS, XOM, and CVX. The defensive sleeve includes JNJ, PG, KO, PEP, and WMT, while diversifiers are captured through GLD, TLT, XLU, and VNQ. SPY serves as the passive equity benchmark throughout.

4.2 Data Sources

The pipeline supports both a free baseline configuration and a richer research configuration. Daily prices and volumes are sourced via `yfinance`. When a `FRED_API_KEY` is available, macro data are drawn directly from FRED; in its absence, the system falls back to BLS and U.S. Treasury Fiscal Data. SEC company facts are incorporated when `SEC_USER_AGENT` is set, enabling a light fundamentals layer without requiring a proprietary data subscription.

4.3 Macro Regime Inputs

When available, the macro sleeve includes rates and labor-market variables such as the 10-year yield, 2-year yield, federal funds rate, unemployment rate, and term spread. The updated research design also incorporates stress-sensitive FRED series including the VIX, the ICE BofA high-yield option-adjusted spread, and a dollar-index proxy. These variables do not form a standalone alpha book. Instead, they help create a macro

regime belief that is blended with the HMM regime estimate and provide additional context for risk-off states.

4.4 Uncertainty and Disagreement Features

One of the most distinctive methodological additions is that the controller is no longer conditioned only on point estimates. The updated state construction also tracks uncertainty proxies derived from the alpha and regime stack. The main examples are alpha-sleeve dispersion, regime-belief entropy, and rolling instability in recent information coefficients. Intuitively, these features measure three different kinds of uncertainty: disagreement across signals, ambiguity in the inferred market state, and instability in the recent mapping from scores to realized returns.

This matters because a sequential controller should not react the same way when the alpha opportunity looks strong and coherent as when the same opportunity is accompanied by high model disagreement or unstable recent signal quality. In that sense, the control layer is not just regime-aware; it is also uncertainty-aware.

4.5 Information-Timing Contract

One of the most important research upgrades in the current version is that the pipeline now makes its feature-availability assumptions explicit. Macro inputs are shifted forward by a configurable number of trading days before they are allowed into the backtest. This is still weaker than a true real-time vintage database, because it does not model revisions, but it is materially stronger than using the final macro history with no delay assumption at all.

Fundamental inputs are handled more conservatively. The SEC quality signal is still a static cross-sectional research prior rather than a fully point-in-time fundamentals panel. For that reason the research engine now supports a strict timing mode in which the SEC quality sleeve is disabled and macro inputs are forced to use the largest configured lag. The point of this design is not to pretend that point-in-time fundamentals are already solved, but rather to make the timing caveat measurable and visible in the evaluation workflow.

5 Methodology

5.1 Problem Formulation

Let $r_{t+1} \in \mathbb{R}^N$ denote next-period asset returns for a universe of N tradable assets. At time t, the system observes a feature state $x_t \in \mathcal{X}$ constructed from cross-sectional factor values, relative-value spreads, volatility forecasts, regime beliefs, and macro inputs. The alpha layer maps these features into an expected-return score vector $\alpha_t \in \mathbb{R}^N$ and a confidence vector $c_t \in \mathbb{R}^N$.

The control layer then chooses a portfolio allocation w_t, a cash allocation, and a hedge intensity h_t. The one-step portfolio objective is not written as a pure return maximization problem. Instead, it is implicitly multi-objective: maximize long-run wealth while limiting drawdown, volatility, and unnecessary turnover. This is the reason the project separates alpha estimation from control: the two tasks operate on different time scales and carry different inductive biases. Importantly, the RL layer does not forecast r_{t+1} directly; it maps a state summarizing alpha quality and risk conditions into exposure, overlay, and protection decisions.

5.2 Cross-Sectional Factor Model

The factor sleeve combines four z-scored components: momentum, a value proxy, a quality proxy, and a low-volatility tilt. Letting m_i, v_i, q_i, and ℓ_i denote the standardized scores for asset i, the composite factor

score is

$$f_i = 0.30m_i + 0.25v_i + 0.25q_i + 0.20\ell_i.$$

This choice reflects a balanced, multi-style factor book rather than a single directional bet.

5.3 Pairs Trading

For candidate pairs, a hedge ratio β is estimated by OLS and the spread

$$s_t = P_t^{(1)} - \beta P_t^{(2)}$$

is normalized into a z-score

$$z_t = \frac{s_t - \mu_s}{\sigma_s + \varepsilon}.$$

Large positive z-scores correspond to short-spread trades and large negative z-scores correspond to long-spread trades.

5.4 Volatility and Regime Models

Each name receives a GARCH(1,1) forecast

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $\omega > 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$ are the intercept, ARCH, and GARCH coefficients respectively. This conditional variance estimate is used for confidence scaling and risk awareness. Separately, a two-state HMM produces a bull-versus-bear regime belief. The final regime signal is

$$\text{RegimeBelief}_t = 0.75 \cdot \text{HMMBelief}_t + 0.25 \cdot \text{MacroBelief}_t.$$

5.5 Adaptive Alpha Combination

Let $s_{i,t}^{(k)}$ denote the signal from source k for asset i at time t. The expected return score is formed as

$$\alpha_{i,t} = \sum_k w_t^{(k)} s_{i,t}^{(k)},$$

where the source weights $w_t^{(k)}$ adapt over time based on recent realized signal quality. In the implementation, this is proxied by recent Spearman information coefficients between source scores and realized returns.

This adaptive weighting is important because the constituent alpha sleeves are not equally informative across all market states. Pairs signals can be weak in strongly trending markets, deep nonlinear forecasts can degrade when regime structure shifts, and factor signals can vary in efficacy depending on the dispersion of the cross-section. The adaptive combiner therefore behaves like a simple online model-selection layer.

5.6 Portfolio Construction RL

The portfolio RL state summarizes the current cross-sectional alpha opportunity, recent realized portfolio volatility, the prevailing regime belief, and an uncertainty score built from alpha dispersion, regime entropy, and recent IC instability. Crucially, the policy is *factor-anchored*. The target book is

$$w_t^{\text{target}} = 0.90\, w_t^{\text{factor}} + 0.10\, w_t^{\text{stabilizer}},$$

where the stabilizer is a long-only minimum-variance-style portfolio estimated from a Ledoit–Wolf shrinkage covariance matrix.

If the RL agent chooses invested fraction b_t and active overlay size τ_t, the final invested portfolio is approximately

$$w_t = b_t \left[(1-\tau_t)w_t^{\text{target}} + \tau_t w_t^{\text{satellite}}\right].$$

The satellite sleeve is not a fully independent book; it tilts the factor target toward the strongest alpha names. This keeps the full pipeline close to the working factor engine while still letting RL modulate aggressiveness.

Between the alpha combiner and the RL policy, the code now inserts a constrained intermediate allocator. The optimizer is long-only and solves a small regularized problem of the form

$$\min_{w} \quad \lambda_{\text{risk}} w^\top \Sigma_t w - \lambda_\alpha \alpha_t^\top w + \lambda_{\text{anchor}} \|w - w_t^{\text{target}}\|_2^2 + \lambda_{\text{turn}} \|w - w_{t-1}\|_2^2$$

subject to $w_i \geq 0$, $\sum_i w_i = 1$, per-asset caps, and simple group concentration limits. This layer is architecturally important because it makes the separation of concerns sharper: alpha expresses expected return, the optimizer expresses institutional portfolio constraints, and RL governs when to lean into or away from that target book. This intermediate layer is not only an architectural convenience; it reflects the practical fact that many real portfolio systems mediate between forecast signals and final positions through a constrained allocator.

5.7 Reward Design

The portfolio RL update uses a differential Sharpe-style reward based on the realized portfolio return relative to a recent rolling distribution of returns. In compact form,

$$R_t^{\text{portfolio}} = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t + \varepsilon},$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are running estimates of recent mean return and volatility. This means the agent is not rewarded solely for taking more risk in absolute terms; it is rewarded for improving return relative to the recent volatility environment.

The hedge agent uses an asymmetric realized-return reward in which losses are penalized more heavily than gains are rewarded. A stylized form is

$$R_t^{\text{hedge}} = r_t \mathbf{1}(r_t \geq 0) + \lambda_{\text{loss}} r_t \mathbf{1}(r_t < 0), \qquad \lambda_{\text{loss}} > 1.$$

That asymmetry is a deliberate design choice: it pushes the hedge layer toward preserving capital in stress rather than maximizing average carry.

The research runner is now also designed to support reward-function ablations. In addition to the default differential-Sharpe-style objective, the evaluation protocol can compare raw-return and Sortino-style reward formulations. This matters because reward design is one of the least observable but most important degrees of freedom in RL-for-finance systems. Because reward specification changes the controller's operating point, it should be treated as a methodological degree of freedom rather than as a neutral implementation detail.

5.8 No-Trade Band and Hedging

To reduce turnover drag, a rebalance band suppresses small changes in target weights. When only part of the portfolio is changed, the active weights are rescaled to preserve the desired capital budget.

The hedging RL layer observes drawdown, volatility regime, and recent momentum. In the archived experiment reported here, it chooses one of four hedge intensities. The total one-period return can be written schematically as

$$r_t^{\text{total}} = r_t^{\text{risky}}(1 - \phi h_t) + r_t^{\text{cash}} + r_t^{\text{hedge}},$$

where h_t is the hedge fraction and r_t^{hedge} includes both carry cost and a convex stress payoff approximation.

Although stylized, this formulation captures the intended economics of a protection sleeve: hedging should be mildly costly in calm periods, increasingly helpful in volatile or negative markets, and most valuable when drawdowns and adverse momentum coincide. The reported hedge sleeve should therefore be interpreted as a tractable convex-loss proxy, not as a market-consistent options strategy.

5.9 Transaction Costs and Execution Realism

The backtest now includes a three-part execution-cost approximation. For each rebalance, the strategy pays a fixed basis-point fee on turnover, an additional penalty proportional to turnover times realized volatility, and a size penalty linked to the largest one-name weight change. In symbols,

$$\text{TC}_t = c_0 \cdot \text{turnover}_t + c_1 \cdot \text{turnover}_t \cdot \hat{\sigma}_t + c_2 \cdot \max_i |\Delta w_{i,t}|.$$

This is not a full execution simulator, but it pushes the daily allocator away from unrealistic churn and makes the rebalance band economically meaningful rather than purely cosmetic.

6 Experimental Design

The archived bundle used in the paper contains approximately 1,250 daily observations from late March 2021 through late March 2026. The reported main split trains the upstream statistical components on the initial segment through March 23, 2023 and then evaluates the walk-forward controller from March 23, 2023 to March 24, 2026. The manuscript is tied to this frozen archived bundle, used as the latest complete saved research run for the current configuration, even though the live codebase may already contain extensions not exercised by the reported evidence. This does not eliminate prior exploratory iteration, but it does make the reported evidence bundle explicit and reproducible. Within the archived suite itself, action ladders, optimizer settings, reward choices, and comparison baselines are then held fixed across ablations, split-fraction tests, and rolling-window robustness slices.

Within that archived run, the LSTM sleeve is first trained on the initial window, after which the full pipeline is rolled forward one trading day at a time with the configuration held fixed during evaluation.

At each date, the pipeline executes the following sequence. First, factor, pairs, GARCH, HMM, macro, and LSTM features are generated from the available data. These are combined into expected-return scores and confidence levels via the adaptive combiner. The portfolio RL agent then selects an exposure bucket and overlay size, after which the hedging agent independently chooses the hedge intensity. Transaction costs, cash carry, and hedge payoffs are applied to arrive at the realized portfolio return, which is finally used to update both RL value tables before advancing to the next trading day.

The comparison set now goes beyond passive and naive portfolio baselines. In addition to SPY buy-and-hold, equal-weight, and the factor-only benchmark, the research protocol includes a volatility-targeting overlay on the factor book, a drawdown-based deleveraging rule, and an end-to-end PPO baseline that receives the same broad information set as the modular architecture. These additions are essential for the paper's research questions because they test whether RL adds value over simple heuristics and whether modular control is preferable to a monolithic learning policy.

6.1 Benchmark Logic

The benchmark set is chosen to isolate different sources of value. SPY measures whether the pipeline adds value relative to a standard passive equity exposure. Equal weight tests whether the broadened universe alone produces useful diversification benefits. The factor-only benchmark isolates the pure alpha layer without any RL overlay. Comparing the full pipeline against the factor-only benchmark is particularly important because it reveals whether RL is adding value as a controller or merely acting as a drag on a strong stock-selection engine.

The newer comparison baselines sharpen that logic further. The volatility targeting baseline asks whether most of the observed improvement can be matched by simple inverse-vol scaling. The drawdown-based deleveraging baseline asks whether a threshold rule can replicate the hedge sleeve's protection. The end-to-end PPO baseline addresses the paper's most distinctive architectural question: if one hands the same state information to a monolithic policy, does it outperform a hybrid design in which finance-specific modules produce the target book and RL only controls how aggressively to express it? These baselines are deliberately strong because they represent the kinds of simple overlays a practitioner would try before accepting the complexity of RL.

6.2 Transaction Costs and Carry

The backtest includes stylized transaction costs together with daily risk-free carry on uninvested cash. These assumptions remain deliberately lightweight, but they are sufficient to force the allocator to confront a meaningful trade-off between responsiveness and trading friction. The no-trade band is best understood in this context: it is not just an implementation convenience, but a structural attempt to reduce noise trading in a daily-rebalanced system.

6.3 Ablation and Robustness Engine

To move the project beyond a single realized path, the codebase now contains a dedicated research runner, `quant_research.py`, which exports structured CSV, JSON, and figure artifacts. The evaluation engine runs several classes of experiments:

1. component ablations comparing factor-only, alpha stack without RL, alpha plus portfolio RL, alpha plus hedge RL, and the full pipeline;
2. multiple train/test split fractions to test dependence on a single partition choice;
3. rolling-window walk-forward evaluations, now expanded to overlapping windows rather than only one or two coarse slices, to assess subperiod stability;
4. sensitivity sweeps over transaction costs, rebalance bands, hedge ladders, and macro lag assumptions; and
5. reward-function ablations, block-bootstrap confidence intervals, and a strict timing-discipline run that disables the static SEC prior and forces lagged macro inputs.

This is one of the most important additions from a research perspective. The main claim of the project is architectural: RL contributes most when it acts as a controller layered on top of financially motivated signals. That claim is much harder to assess from a single equity curve than from a battery of ablations and robustness checks.

7 Evaluation Metrics

The reporting suite includes annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, and return-distribution diagnostics including a daily 5% Value-at-Risk estimate.

If $\{r_t\}_{t=1}^{T}$ denotes daily portfolio returns and r_f is the annual risk-free rate, then the annualized Sharpe ratio is

$$\text{Sharpe} = \frac{252\,\bar{r} - r_f}{\sqrt{252}\,\hat{\sigma}(r) + \varepsilon}.$$

The Sortino ratio replaces total volatility with downside volatility, while the Calmar ratio scales annualized return by the magnitude of maximum drawdown. Each metric emphasizes a different aspect of the return distribution, which matters in this project because the pipeline is explicitly designed to reshape tail risk, not merely to maximize mean return. Table 3 focuses on the primary comparison metrics used in the main strategy table: return, volatility, Sharpe, maximum drawdown, and Calmar. Sortino and Value-at-Risk are retained as secondary diagnostics in the text, reward-ablation evidence, and archived research artifacts.

In addition, the research runner emits regime-conditional summaries of portfolio action, hedge ratio, cash weight, turnover, transaction-cost drag, and realized returns. These diagnostics help evaluate whether the controller behaves differently in bullish, neutral, and bearish states in a way that is consistent with the design intent of the policy.

For statistical reliability, the updated evaluation design also includes block-bootstrap confidence intervals on daily-return paths. The motivation is to preserve some of the short-horizon dependence structure in returns while still providing uncertainty bands for quantities such as annualized return, Sharpe, Calmar, and pairwise performance differences versus benchmark strategies.

8 Results

This section moves from the main walk-forward evidence to ablation, comparisons against rule-based overlays and end-to-end PPO, rolling robustness slices, and regime-conditioned behavior. The organizing question throughout is whether the data support the narrower claim that RL helps more as a controller than as a predictor.

8.1 Main Result

The archived research run used for the paper's main evidence uses a test window from March 23, 2023 to March 24, 2026. The summary metrics are shown in Table 3. This is still only one realized path, but it is the same saved checkpoint that also feeds the ablation, robustness, and significance artifacts discussed in the remainder of the paper.

Table 3: Performance summary on the walk-forward test window (23 Mar 2023 – 24 Mar 2026). Annualized figures. Max DD denotes maximum peak-to-trough drawdown; Calmar is annualized return divided by |Max DD|.

Strategy	**Return**	**Volatility**	**Sharpe**	**Max DD**	**Calmar**
Full pipeline	17.2%	11.1%	1.23	−10.2%	1.68
SPY	17.9%	15.7%	0.92	−18.8%	0.96
Equal-weight	22.8%	11.7%	1.65	−14.8%	1.54
Factor-only benchmark	33.1%	15.3%	1.94	−16.5%	2.01
Vol-target	28.9%	13.2%	1.92	−14.8%	1.95
DD-Delever	30.2%	13.7%	1.95	−14.9%	2.02
E2E RL (PPO)	9.1%	9.1%	0.62	−14.0%	0.65

The main result is partly negative and partly favorable, which is exactly what makes it more credible than the earlier best-case narrative.

Relative to SPY, the full pipeline is almost flat on annualized return, 17.2% versus 17.9%, while materially improving volatility, Sharpe, and drawdown. The passive-benchmark comparison therefore supports the claim that the controller can reshape downside risk in a useful way. The relatively strong equal-weight result is also informative: some of the benefit in this universe comes from broad diversification across growth, defensive, and diversifier sleeves rather than from RL control alone.

Relative to the factor-only benchmark, the conclusion is less favorable. The full pipeline is far below the factor core on both return and Sharpe, so the reported evidence does not support any stronger claim that RL preserves most of the upside of the alpha engine.

Relative to the strongest rule-based overlays, the result is also negative. Both volatility targeting and drawdown-based deleveraging outperform the full pipeline on return- and Sharpe-based comparisons in the archived sample.

Relative to the end-to-end PPO baseline, the result is directionally favorable. The modular pipeline substantially outperforms PPO on archived-sample point estimates and avoids the much weaker return profile of the monolithic policy.

Implication for RQ1 and RQ2. The main table alone already suggests a qualified positive-negative result: favorable for modular RL relative to PPO and passive exposure, but not favorable for RL relative to the strongest non-RL overlays.

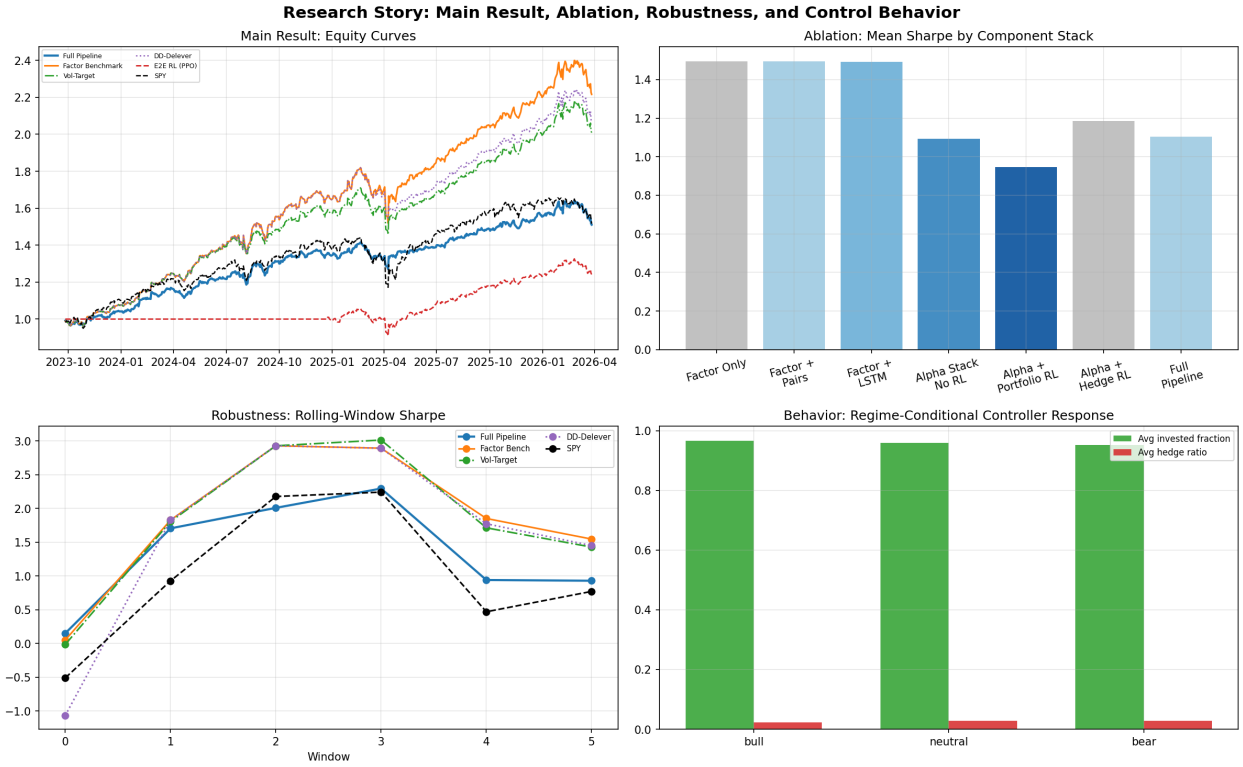

Figure 2: Compact workshop-style summary figure generated by the robustness engine. The four panels show the main equity-curve result, ablation evidence, rolling-window robustness, and regime-conditional controller behavior. The figure makes clear that the strongest raw-performance outcomes in the archived bundle come from the factor-only benchmark and the strongest simple overlays, not from the full RL stack, while also showing that the modular RL stack remains visibly above the end-to-end PPO baseline.

8.2 Ablation Result

The most important workshop-level question is not whether the full system does well on one path, but whether the component layering supports the paper's architectural claim. For that reason the research runner now emits an ablation summary comparing the factor-only benchmark, the alpha stack without RL, alpha plus portfolio RL, alpha plus hedge RL, and the full pipeline across multiple split settings.

Table 4: Ablation summary across train/test split settings.

Component	Return	Vol.	Sharpe	Max DD	Calmar
Factor-only benchmark	21.7%	12.2%	1.50	−14.0%	1.55
Factor + pairs	21.7%	12.2%	1.50	−13.9%	1.56
Factor + LSTM	21.6%	12.2%	1.49	−13.9%	1.55
Alpha stack (no RL)	17.7%	13.0%	1.09	−14.5%	1.22
Alpha + portfolio RL	14.7%	11.9%	0.95	−13.5%	1.09
Alpha + hedge RL	18.0%	12.3%	1.18	−13.1%	1.38
Full pipeline	15.7%	11.1%	1.10	−11.3%	1.43

Table 5: Rolling-window robustness summary for the full pipeline.

Statistic	Value
Rolling windows	6
Median full-pipeline Sharpe	1.32
Median full-pipeline Calmar	2.55
Median full-pipeline Max DD	-7.2%
Frac. windows full beats factor-only benchmark on drawdown	67%
Frac. windows full beats factor-only benchmark on Calmar	17%
Frac. windows full beats SPY on Sharpe	83%
Frac. windows full beats SPY on Calmar	100%

Table 6: Block-bootstrap significance for full pipeline versus baselines (Sharpe deltas).

Comparison	Delta Sharpe	95% CI	p-value
Full – SPY	0.31	[-0.47, 1.01]	0.455
Full – Factor-only benchmark	-0.70	[-1.34, -0.26]	0.000
Full – Vol-target	-0.69	[-1.26, -0.22]	0.005
Full – DD-Delever	-0.72	[-1.27, -0.22]	0.010
Full – E2E RL (PPO)	0.62	[-0.39, 1.50]	0.280

The generated ablation table sharpens the story considerably. Averaged across split settings, the factor-only benchmark remains the strongest pure alpha expression, with mean return of 21.7% and mean Sharpe of 1.50. The pairs and LSTM sleeves add very little incremental value on top of that base. This is a useful pruning result: on top of the factor core in the current archived evaluation bundle, the pairs and LSTM sleeves do not add enough incremental value to justify their complexity. The full pipeline is materially weaker on raw upside, at 15.7% mean return and 1.10 mean Sharpe, but it posts the best mean drawdown profile in the table at -11.3% and the lowest mean volatility at 11.1%. The hedge-only ablation is the strongest intermediate RL configuration, with mean Sharpe 1.18 and mean drawdown -13.1%, while portfolio RL without the hedge sleeve is the weakest controller variant, at mean Sharpe 0.95. This pattern is informative: exposure control on its own can under-express the alpha engine, and most of the RL contribution appears to come through protection and joint portfolio-plus-hedge coordination rather than through aggressive risk taking.

Implication for RQ3. Supported. The ablation evidence points much more strongly to the protection sleeve, and to the joint portfolio-and-hedge design, than to portfolio RL alone.

8.3 Rule-Based and End-to-End Baseline Evidence

The next layer of evidence asks whether the RL controller is doing something meaningfully different from simple heuristics. On this archived run, the volatility-targeting and drawdown-based deleveraging baselines are deliberately strong comparators: both deliver around 29–30% annualized return with Sharpe ratios near 1.93–1.95, well above the full pipeline's 1.23. The bootstrap significance table makes the same point more formally. Relative to vol-target and drawdown-based deleveraging, the full pipeline's Sharpe deficit is around 0.69 to 0.72, and the 95% confidence intervals remain below zero. The significance evidence is therefore asymmetric: the full pipeline's Sharpe deficits relative to the factor-only benchmark and the strongest rule-based overlays survive bootstrap uncertainty, while its apparent advantage over SPY and PPO

does not reach comparable statistical strength in this archived sample.

In the archived sample, RQ2 is not supported. The evidence in this manuscript does not support the claim that RL adds value over the strongest simple rule-based overlays on Sharpe or return. RQ1 receives stronger directional support. The end-to-end PPO baseline reaches only 9.1% annualized return and 0.62 Sharpe, so the modular architecture substantially outperforms PPO on point estimates, even though the bootstrap-based Sharpe delta is not statistically decisive at conventional levels. Here point estimates refer to the archived sample's observed annualized return, Sharpe, and drawdown metrics before bootstrap uncertainty qualification. That is still an important result: the data support RL as a control layer much more strongly than RL as a monolithic trading policy.

Implication for RQ1. Supported directionally: modular RL is materially better than end-to-end PPO in point estimates, but the manuscript does not claim a statistically decisive Sharpe advantage.

Implication for RQ2. Not supported in the archived sample: the strongest simple non-RL overlays outperform the RL controller on return and Sharpe.

8.4 Reward-Function Sensitivity

Reward design affects where the controller lands on the risk-return frontier, but it does not overturn the broader architectural story. Across the full pipeline reward-function ablation, the raw-return reward variant delivers the highest annualized return at 17.7% and the highest Sharpe at 1.26, while the Sortino-style reward produces the best drawdown and Calmar profile, with maximum drawdown improving to -9.4% and Calmar rising to 1.78. The default differential-Sharpe reward is therefore not uniquely optimal; it is simply one reasonable compromise. This is a useful methodological reminder that reward choice materially shifts the policy's operating point even when the state and action spaces are held fixed.

Table 7: Reward-function ablation for the full pipeline on the archived bundle.

Reward	Return	Sharpe	Max DD	Calmar
Differential Sharpe	16.4%	1.18	−11.5%	1.42
Return	17.7%	1.26	−10.2%	1.74
Sortino	16.8%	1.23	−9.4%	1.78
Mean-Variance	16.5%	1.18	−9.5%	1.73

The archived reward-ablation evidence is therefore specific to the modular full pipeline. A comparable PPO reward sweep is not treated as primary evidence in this manuscript because the archived suite did not preserve an equally complete reward-by-split comparison grid for PPO, and the smaller set of PPO reward runs is materially noisier because each trial depends on fresh stochastic policy training.

Implication for methodology. Reward choice matters substantially for where the controller lands on the risk-return frontier, but it does not overturn the paper's main qualitative result: reward tuning changes the controller's operating point on the risk-return frontier without overturning the broader qualitative ranking in which the factor-only benchmark and the strongest simple overlays remain stronger on return- and Sharpe-based comparisons.

8.5 Robustness Across Windows

Single-path backtests can be persuasive but are rarely sufficient. The research runner therefore evaluates the full pipeline across rolling windows and then summarizes median Sharpe, median Calmar, drawdown

dispersion, and benchmark beat fractions. In the archived run, the median full-pipeline Sharpe across rolling windows is 1.32, the median Calmar is 2.55, and the median maximum drawdown is -7.2%. The full pipeline beats SPY on Sharpe in 83% of windows and on Calmar in 100% of windows. Relative to the factor-only benchmark, however, the picture is mixed: the full pipeline posts a better drawdown in 67% of windows but a better Calmar in only 17% of them. That nuance is exactly why the rolling-window engine matters. The controller appears structurally valuable for downside reshaping relative to passive exposure, but not yet structurally dominant over the strongest finance-first benchmark. These windows still come from one historical sample and should therefore be interpreted as internal robustness slices rather than independent replications.

The same evidence also helps calibrate the paper's statistical tone. The bootstrap significance summary shows that the full pipeline's Sharpe advantage over SPY is positive in point estimate but not statistically decisive, while its Sharpe disadvantage relative to the factor-only benchmark and the two rule-based overlays is large enough to survive block-bootstrap uncertainty bands. The appropriate claim, then, is not that robustness is "won," but that the architecture now supports a falsifiable and transparent robustness protocol.

Implication for the overall claim. The rolling evidence is favorable for downside reshaping relative to SPY, but only mixed relative to the factor-only benchmark and the strongest rule-based alternatives.

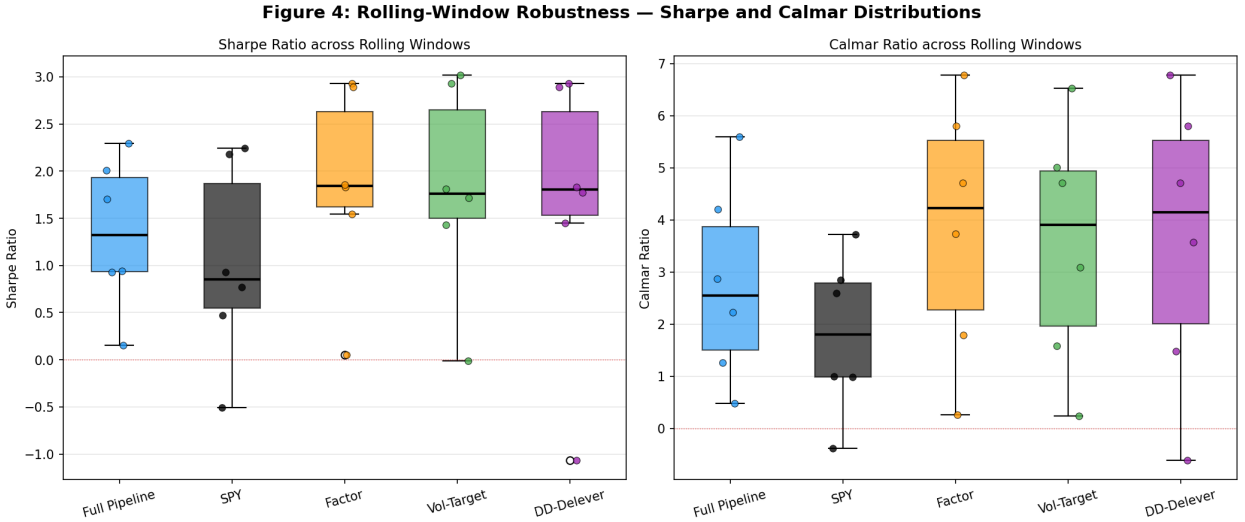

Figure 3: Rolling-window robustness from the archived March 2026 experiment bundle. The full pipeline is consistently stronger than SPY on Calmar and usually stronger on Sharpe, but it remains below the factor-only benchmark and the best rule-based overlays on median upside-oriented metrics. These rolling windows are internal robustness slices from one archived historical sample rather than independent replications.

8.6 Behavioral and Regime Evidence

The behavioral question is whether the controller is acting like a sequential decision system. Figure 2 provides the main summary used in the paper, while supplementary alpha and execution diagnostics are reported in Appendix Figures 4 and 5. The regime summaries show that the controller is not degenerate: it carries nonzero hedge ratios, nontrivial cash weights, and changing turnover profiles rather than collapsing to an always-in, never-hedged policy. At the same time, the archived baseline split shows only modest regime separation. In the representative full-pipeline split, average portfolio action declines from roughly 2.08 in bull states to 1.26 in bear states, and average cash weight rises from about 6.6% to 10.6%, but hedge intensity is comparatively flat. That is a constructive research finding rather than a failure. The controller is reacting,

but mostly through exposure throttling rather than through a strongly state-contingent hedge sleeve. This behavioral pattern is consistent with the ablation evidence that the protection sleeve matters, but that the present hedge policy may still be under-expressive relative to its intended role. This is exactly the kind of behavioral gap that motivates the paper's forward-looking interest in implied-volatility-conditioned and option-style hedging overlays.

Implication for RQ3. Supported, but only partially. The behavioral evidence is consistent with hedge-related control mattering more than portfolio RL alone, while also suggesting that the hedge policy remains less state-contingent than intended.

9 Discussion

The experimental results support three conclusions of particular methodological importance.

First, the data and alpha side of the pipeline is doing real work. The factor-only benchmark is strong, and equal-weight also performs reasonably well, which suggests that the diversified universe and the factor core contribute most of the signal, while the non-factor sleeves currently add limited incremental value in the archived evaluation. The relatively strong equal-weight result is itself informative: part of the observed performance comes from broadening the universe and diversifying across sectors and defensive sleeves, not only from the more elaborate control machinery. In the current archived evaluation, the paper's strongest alpha evidence comes from the factor core, not from the auxiliary pairs or LSTM sleeves.

Second, the factor-anchored RL design is helpful in a narrower sense than simple benchmark outperformance. The ablation, rolling-window, and significance results all suggest that the full pipeline's main contribution is downside-risk transformation rather than universal metric dominance. The factor-only benchmark remains the strongest pure alpha engine, and the simple vol-target and drawdown-based deleveraging overlays are stronger than the RL controller on this sample's primary return- and Sharpe-based metrics.

This distinction is central. If the RL layer were treated as a second stock selector, it would risk duplicating the alpha problem with a less structured model and less data efficiency. By anchoring the policy to the factor book, the pipeline preserves interpretability: alpha determines *what* to own, while RL primarily determines *how much* to own and *how much protection* to carry.

What did not work is equally important. RL did not beat the strongest rule-based overlays on Sharpe or return in the archived sample. Portfolio RL on its own underperformed and could under-express the alpha engine. The pairs and LSTM sleeves added little incremental value beyond the factor core. Finally, the hedge response appears flatter across regimes than the intended design would prefer. These are useful negative results because they narrow where future improvements should be sought. For a control-layer study, these failures are not incidental; they are precisely what helps delimit where RL is and is not adding value.

The results also suggest a useful practical lesson. Improvements in data quality and signal specification are likely to matter more from this point onward than adding more control complexity. The current architecture is already rich enough to express nontrivial risk-management behavior, but the rolling-window evidence shows that the factor-only benchmark and cheap rule-based overlays still dominate the full pipeline on upside-oriented metrics. Future gains will likely come from more realistic point-in-time fundamentals, true macro vintage data, cross-asset regime features, explicit uncertainty conditioning, stronger statistical validation, and richer hedging instruments with implied-volatility information.

9.1 When Practitioners Should Use This Architecture

The archived evidence suggests a fairly specific use case for factor-anchored RL control rather than a universal recipe. The architecture is most defensible when downside protection matters more than raw return maximization, when interpretability matters for institutional adoption or model governance, when a credible finance-first alpha engine already exists, and when the research or production setting can support daily rebalancing together with periodic RL training and monitoring.

Simpler alternatives are often sufficient in other settings. If the primary goal is pure Sharpe maximization on this kind of daily equity universe, the archived bundle suggests that the factor-only benchmark and simple overlays such as volatility targeting or drawdown-based deleveraging are harder to beat. If computational budgets are tight, rule-based overlays are also easier to train, debug, and explain. And if the alpha layer is weak, RL is unlikely to rescue the system: the controller can reshape risk, but it cannot compensate for the absence of robust signal quality.

The practical contribution is therefore not "always use RL." It is narrower: when RL is appropriate in finance, it appears more defensible as a controller over a strong signal engine than as a replacement for prediction itself.

10 Threats to Validity and Future Work

The implementation remains research code, and several caveats apply. Macro inputs are now shifted by a configurable lag, but they are not yet full vintage series with revision-aware histories. SEC enrichment can be used as a static cross-sectional prior, but it still does not constitute a true point-in-time fundamentals database. Transaction costs are more realistic than before, yet they remain stylized rather than venue-specific. The hedge sleeve in the archived experiment approximates convex protection without live options data, and the robustness engine still depends on one underlying historical sample even when it slices that sample in multiple ways.

Several additional threats deserve explicit acknowledgement. First, the design has been iterated by observing backtest behavior, which raises the possibility of researcher degrees of freedom even when each individual change is motivated. Second, the traded universe is broad enough to be interesting but still narrow relative to institutional equity universes, so the results may depend partly on the selected names. Third, daily data are appropriate for a medium-horizon allocation study but cannot validate intraday execution claims. Fourth, the hedge sleeve remains an approximation to convex protection rather than a true options implementation. Finally, the project deliberately leans on free or public data sources, which improves accessibility but necessarily limits point-in-time fidelity and market microstructure realism.

A further caveat is benchmark fairness: although the rule-based and PPO baselines are designed to be strong and informative, their exact specification still influences comparative conclusions. Because the evaluation suite reports many metrics and comparator strategies, some apparent wins and losses should also be interpreted with the usual caution around multiple comparisons. External validity is also limited: conclusions drawn from a daily U.S.-equity allocation setting may not transfer unchanged to other asset classes, frequencies, or market microstructure regimes.

The currently reported robustness summary is still based on only six rolling windows drawn from one historical sample. The workshop interpretation should therefore be that the robustness engine is now part of the methodology, not that robustness has been exhaustively established across many independent market environments.

Natural next steps include true filing-date-aware fundamentals, release-vintage macro datasets, cross-asset

regime instruments, and a richer hedge sleeve using implied-volatility-conditioned option overlays such as protective puts, collars, and put spreads. A second important line of follow-up work is empirical pruning: the LSTM sleeve and the pairs sleeve should remain in the architecture only if ablation continues to show incremental value after the expanded robustness pass. Finally, while the evaluation engine now includes bootstrap significance outputs, future work could push further toward formal Sharpe-ratio comparison tests, additional universes, and deeper cross-validation across disjoint market regimes.

11 Conclusion

This paper shows that a financially grounded hybrid architecture can be modular, interpretable, and empirically informative even when its RL layer does not beat the strongest finance-first baselines on top-line performance. Classical quantitative models remain responsible for alpha discovery, while reinforcement learning acts as a sequential risk and allocation controller. The resulting system is not a proof that RL dominates traditional quant methods. Instead, it is a structured research pipeline whose behavior is interpretable, testable, and falsifiable. The main empirical takeaway is accordingly narrow but meaningful: when RL is anchored to a strong finance-first alpha book and evaluated under explicit timing and cost assumptions, it can improve downside control relative to passive exposure and outperform end-to-end RL on point estimates, even if strong factor-first and rule-based baselines still set a higher bar on pure Sharpe and return. Equally importantly, the paper leaves behind a frozen-bundle evaluation protocol that can support future pruning, hedge redesign, and options-based extensions under a consistent empirical standard. In the reported sample, the evidence supports RL as a disciplined control overlay for downside shaping and rejects the stronger claim that RL already surpasses the best finance-first or rule-based alternatives on return and Sharpe.

A Implementation Snapshot

A.1 Archived Experiment Control Ladders

The March 2026 archived experiment bundle used in the main paper results relies on the discrete action ladders below. The live codebase now contains early scaffolding for implied-volatility-conditioned, option-style hedge selection, but those extensions are not yet part of the reported archived evaluation.

Portfolio action	Invested fraction	Active overlay fraction
State 0	82%	5%
State 1	90%	8%
State 2	95%	12%
State 3	98%	16%
State 4	100%	20%

Hedge action	Hedge ratio
State 0	0%
State 1	3%
State 2	8%
State 3	15%

A.2 Research Artifacts

For the archived run discussed in the paper, the research runner writes the artifacts below into `results/20260327_experiments/stock_trading/` alongside the standard pipeline figures:

1. `research_metrics.csv`: tabular output for ablation and robustness runs;
2. `research_ablation_summary.csv`: split-aggregated ablation summaries;
3. `research_robustness_summary.csv`: rolling-window comparison statistics;
4. `research_regime_summary.csv`: regime-conditioned action and return summaries;
5. `research_rolling_references.csv`: rolling benchmark metrics for SPY and the factor-only benchmark;
6. `research_bootstrap_cis.csv`: block-bootstrap confidence intervals for key strategy metrics;
7. `research_bootstrap_significance.csv`: pairwise bootstrap deltas and p-value summaries versus benchmark strategies;
8. `research_summary.json`: serialized configuration and best-per-suite summaries; and
9. `research_paper_tables.tex` together with `pipeline_research_eval.png`: paper-ready summary tables and figure.

Together with the feature-availability contract embedded in each backtest run, these artifacts make it easier to compare results across alternative timing, cost, and control assumptions without manually reconstructing the experimental context after the fact.

A.3 Supplementary Diagnostic Figures

These figures are diagnostic and illustrative; the primary evidentiary claims of the paper rest on Tables 3–6 and Figures 2 and 3. The appendix also intentionally omits the older RL dashboard with embedded performance text, because those annotations reflect an exploratory live run rather than the frozen archived bundle analyzed in the paper.

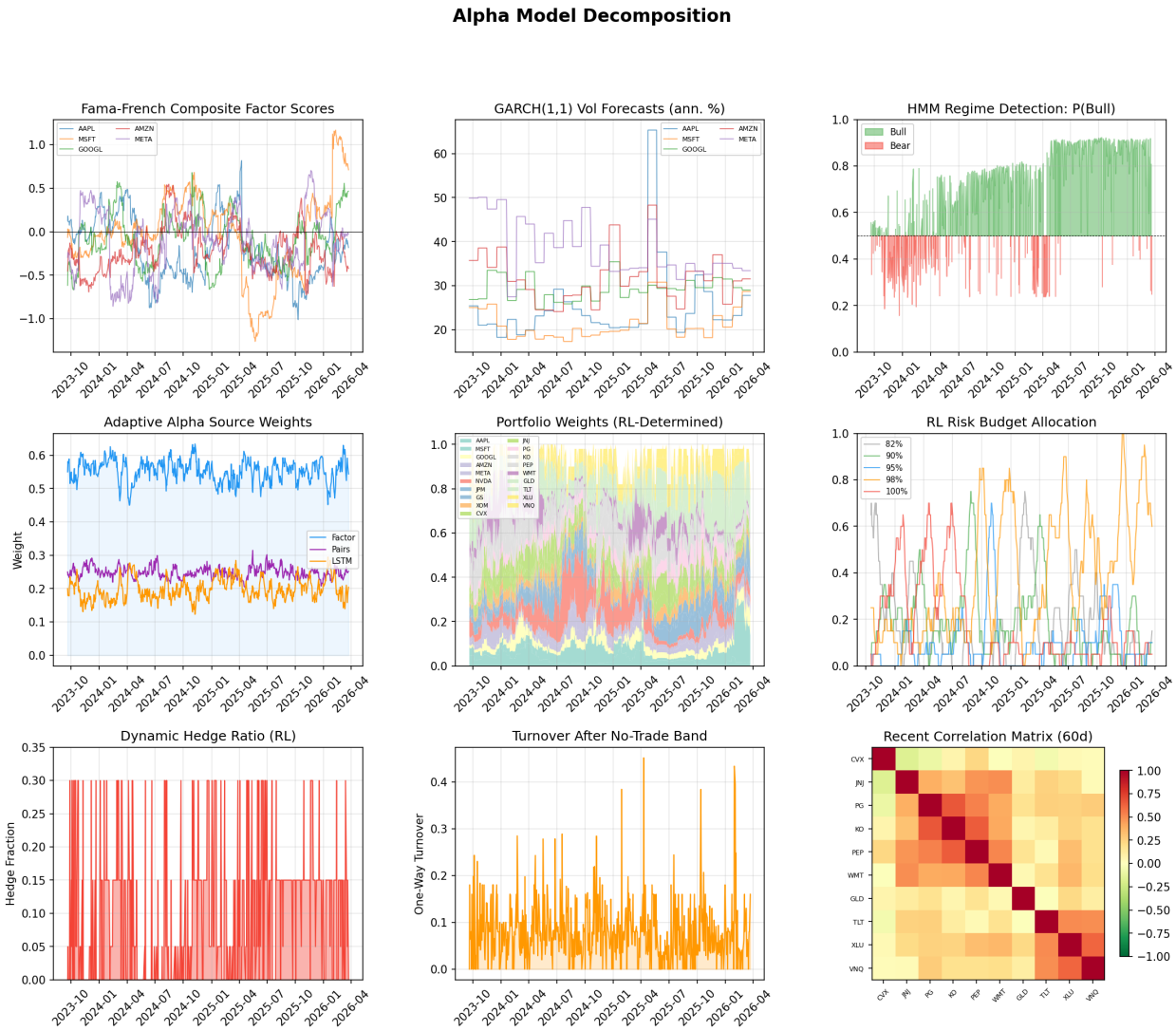

Figure 4: Supplementary alpha diagnostics for the archived bundle. The figure is diagnostic rather than primary evidence and is consistent with the paper's pruning conclusion that the factor core carries most of the demonstrable alpha value in the current evaluation.

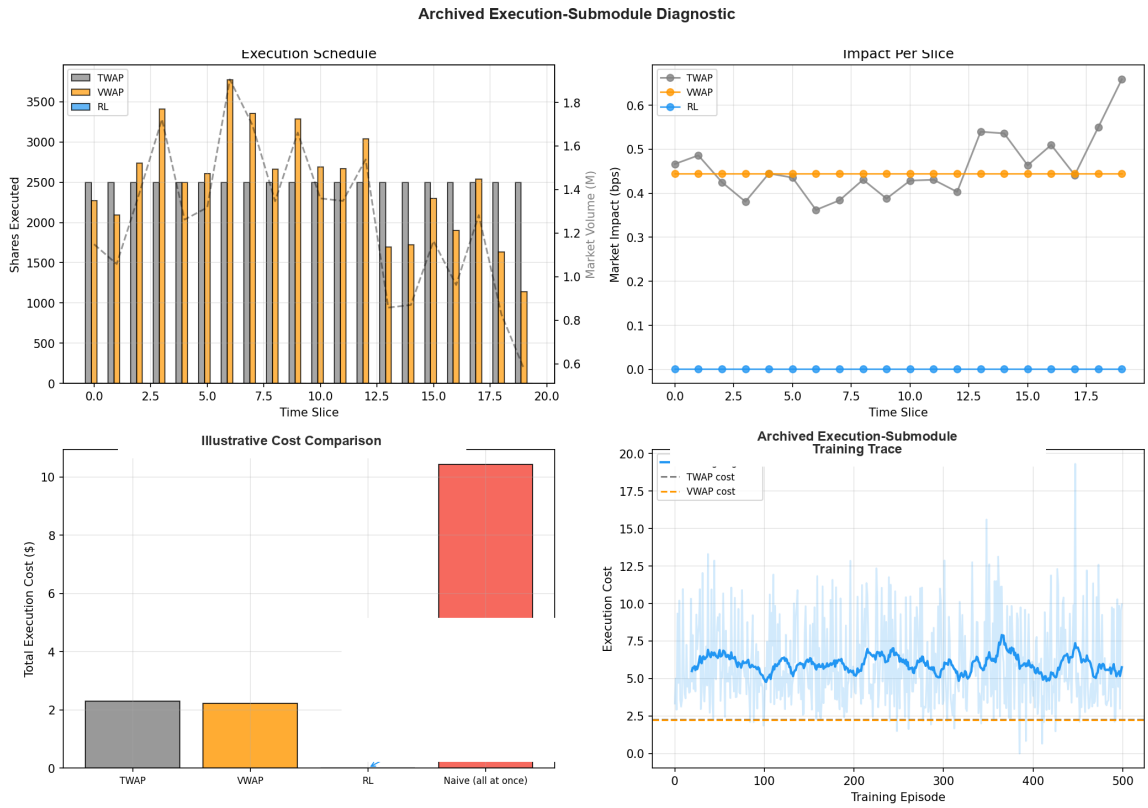

Figure 5: Supplementary execution diagnostic for the archived execution submodule. The figure illustrates turnover management, no-trade band activations, and stylized transaction-cost behavior, but it is not treated as a primary claim about live trading execution realism.

References

[1] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2):5–39, 2000.

[2] H. Buehler, L. Gonon, J. Teichmann, and B. Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019.

[3] T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[4] L. Chen, M. Pelger, and J. Zhu. Deep learning in asset pricing. *SSRN Working Paper 3350138*, revised 2021.

[5] Y. Cong, B. Deng, X. Bao, and H. Dai. AlphaPortfolio: Direct construction through deep reinforcement learning and interpretable representation learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(6):4831–4839, 2021.

[6] Y. Deng, F. Bao, Y. Kong, Z. Ren, and Q. Dai. Deep direct reinforcement learning for financial signal representation and trading. *IEEE Transactions on Neural Networks and Learning Systems*, 28(3):653–664, 2017.

[7] Y. Du, P. N. Kolm, A. Ritter, and T. Wang. Deep reinforcement learning for option replication and hedging. *The Journal of Financial Data Science*, 2(4):44–59, 2020.

[8] R. F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

[9] R. F. Engle and C. W. J. Granger. Co-integration and error correction: representation, estimation, and testing. *Econometrica*, 55(2):251–276, 1987.

[10] E. F. Fama and K. R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.

[11] S. Giglio, D. Xiu, and D. Zhang. Machine learning in asset pricing. *Annual Review of Financial Economics*, 14:337–368, 2022.

[12] S. Gu, B. Kelly, and D. Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

[13] J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.

[14] J. Jander, H. H. Li, M. Schmid, and T. Pohl. Reinforcement learning in finance: From current practice to standards and benchmarks. *SSRN Working Paper 5703882*, 2025.

[15] N. Jegadeesh and S. Titman. Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.

[16] Z. Jiang, D. Xu, and J. Liang. A deep reinforcement learning framework for the financial portfolio management problem. *arXiv preprint arXiv:1706.10059*, 2017.

[17] P. N. Kolm and G. Ritter. Modern perspectives on reinforcement learning in finance. *SSRN Working Paper 3449401*, 2019.

[18] P. N. Kolm and G. Ritter. Dynamic replication and hedging: A reinforcement learning approach. *The Journal of Financial Data Science*, 1(4):159–171, 2019.

[19] J. Moody and M. Saffell. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4):875–889, 2001.

[20] Y. Nevmyvaka, Y. Feng, and M. Kearns. Reinforcement learning for optimized trade execution. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 673–680, 2006.

[21] D. Pacheco Aznar. Portfolio management: A deep distributional reinforcement learning approach. *SSRN Working Paper 4522084*, 2023.

[22] S. Sun, M. Qin, W. Zhang, H. Xia, C. Zong, J. Ying, Y. Xie, L. Zhao, X. Wang, and B. An. TradeMaster: A holistic quantitative trading platform empowered by reinforcement learning. In *Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track*, 2023.

[23] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, second edition, 2018.

[24] Z. Zhang, S. Zohren, and S. Roberts. Deep reinforcement learning for trading. *The Journal of Financial Data Science*, 2(2):25–40, 2020.